Bagged Offset (production)

offset family, scored against the market on 2015-2026, 5 seed(s), from 05f09ed2455d

Contents

Model identity and provenance

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL	MEASURED AGAINST	SEEDS	TEST YEARS	BOOSTING ROUNDS
Bagged Offset (production) offset	the market the same yardstick in all four reports	5 7, 17, 42, 123, 256	12 2015-2026	400 3-fold CV

TRAINED ON	SOURCE COMMIT
2010-2014 then expanding a separate fit per test year, each on 2010 through the year before it -- 12 fits in all	05f09ed2455d clean

These are walk-forward figures, not one model's: for each test year the model was fitted on 2010 through the previous year and then scored that year, so the result concatenates 12 separately fitted models and no test year appears in its own training data. The window expands each year, so the last year's model saw the most data and the first year's the least.

Config digest `sha256:cb9351b6601ede16f58a647a4b844d09ee84c3398155cdc2ee48b917081c41a9`, model view `bagged_offset`, artifacts read from `results/model_report`.

ENVIRONMENT

Component	Version
numpy	2.5.2
pandas	3.0.5
python	3.12.11
xgboost	3.4.1

Deltas in this report use `model_minus_market_log_loss`: race_multinomial_log_loss, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Executive summary

MEAN DELTA VS MARKET	RACES	YEARS BETTER	RACE-LEVEL WIN RATE
-0.004647 negative is better	8,675	11/12	54.3%

Bagged Offset (production) is better than the market by 0.004647 race_multinomial_log_loss on 8,675 races. The direction holds in 11 of 12 evaluated years. Race by race it is ahead 54.3% of the time, which is the more honest figure: a mean delta can be carried by a few races. 1,398 bets were selected under the staking rules below, on retrospective odds that could not have been bet into.

Comparator and evaluation population

The market's own implied probabilities on the same races. Every published report is scored against the market, so the four headline numbers mean the same thing and can be read against one another. For the production model the market is also the baseline by construction -- the booster learns a residual on log(market_prob) -- and for the probit candidates it is the common yardstick rather than the quantity they were designed against; the comparison each was designed against is in the technical appendix.

TEST RACES	YEARS
8,675	12 2015-2026

Splits present: `test`, `train_apparent`, `train_prequential`. Only the forward test split is scored above. The others exist to be compared against it -- an apparent-fit number looks far better than a forward one -- and are never pooled with it.

Out-of-fold margin sources: `market_fallback`, `model`.

One or more years is partial

2026 does not span a full season, so its per-year figures rest on fewer races than the others. Partial years are marked in every table rather than dropped, because dropping them would silently change the evaluation population.

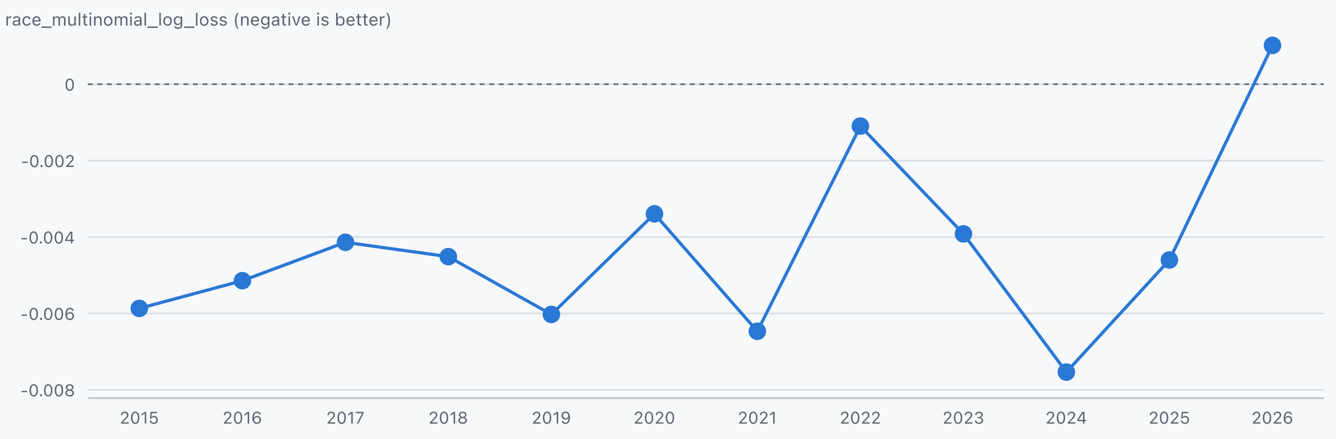
Overall probability performance

races	model log loss	market log loss	model minus market log loss	better race share
8675	2.021454	2.026101	-0.004647	0.543285

One row, pooled over every evaluated race, and measured against the market: `model_minus_market_log_loss` is the model's race log loss minus the market's on the same races, so negative means the model is better. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	market log loss	model minus market log loss	better race share	is partial year
2015	742	1.992850	1.998720	-0.005869	0.549865	no
2016	744	2.094135	2.099277	-0.005142	0.564516	no
2017	772	2.017423	2.021559	-0.004136	0.528497	no
2018	756	2.056990	2.061504	-0.004514	0.525132	no
2019	759	1.926801	1.932827	-0.006026	0.548090	no
2020	799	2.007010	2.010400	-0.003390	0.535670	no
2021	783	2.000193	2.006659	-0.006466	0.565773	no
2022	783	2.050100	2.051195	-0.001095	0.519796	no
2023	796	2.020147	2.024066	-0.003919	0.530151	no
2024	792	2.018210	2.025744	-0.007534	0.565657	no
2025	748	2.052795	2.057397	-0.004601	0.554813	no
2026	*201*	*2.027241*	*2.026224*	*0.001016*	*0.502488*	*yes*



Mean delta by year Mean delta against the market for each evaluated year. Zero is the market.

▼ Table view

Category	race_multinomial_log_loss (negative is better)
2015	-0.005869

Category	race_multinomial_log_loss (negative is better)
2016	-0.005142
2017	-0.004136
2018	-0.004514
2019	-0.006026
2020	-0.00339
2021	-0.006466
2022	-0.001095
2023	-0.003919
2024	-0.007534
2025	-0.004601
2026	0.001016

Ahead of the market in 11 of 12 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

Reliability by pipeline stage, on bins fixed across the whole report so two stages are comparable, pooled from the per-year counts rather than averaged over years. Stages: `market`, `raw_offset`, `temperature`, `market_blend`, `bagged`.

MARKET

Pooled over every test year.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	12,309	38	0.0063	0.0031	0.493
0.01-0.03	18,118	288	0.0171	0.0159	0.929
0.03-0.05	19,584	693	0.0365	0.0354	0.969
0.05-0.07	16,510	980	0.0625	0.0594	0.950
0.07-0.10	11,221	907	0.0874	0.0808	0.925
0.10-0.15	12,867	1,645	0.1223	0.1278	1.045
0.15-0.20	6,866	1,226	0.1727	0.1786	1.034
0.20-0.30	6,211	1,533	0.2418	0.2468	1.021
0.30-0.50	3,019	1,153	0.3653	0.3819	1.046
0.50-1.00	354	212	0.5542	0.5989	1.081

RAW_OFFSET

Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	71,707	316	0.0058	0.0044	0.759
0.01-0.03	90,687	1,502	0.0171	0.0166	0.969
0.03-0.05	100,056	3,811	0.0368	0.0381	1.036
0.05-0.07	73,499	4,447	0.0620	0.0605	0.976
0.07-0.10	53,920	4,526	0.0866	0.0839	0.969
0.10-0.15	60,790	7,787	0.1224	0.1281	1.046
0.15-0.20	33,222	5,789	0.1728	0.1743	1.008
0.20-0.30	32,301	7,735	0.2423	0.2395	0.988
0.30-0.50	16,907	6,191	0.3692	0.3662	0.992
0.50-1.00	2,206	1,271	0.5672	0.5762	1.016

TEMPERATURE

Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	74,865	356	0.0057	0.0048	0.836
0.01-0.03	90,629	1,557	0.0171	0.0172	1.006

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.03-0.05	99,137	3,826	0.0367	0.0386	1.050
0.05-0.07	72,791	4,431	0.0619	0.0609	0.983
0.07-0.10	53,074	4,485	0.0866	0.0845	0.976
0.10-0.15	59,848	7,664	0.1224	0.1281	1.046
0.15-0.20	32,786	5,686	0.1727	0.1734	1.004
0.20-0.30	32,279	7,710	0.2422	0.2389	0.986
0.30-0.50	17,427	6,270	0.3697	0.3598	0.973
0.50-1.00	2,459	1,390	0.5704	0.5653	0.991

MARKET_BLEND

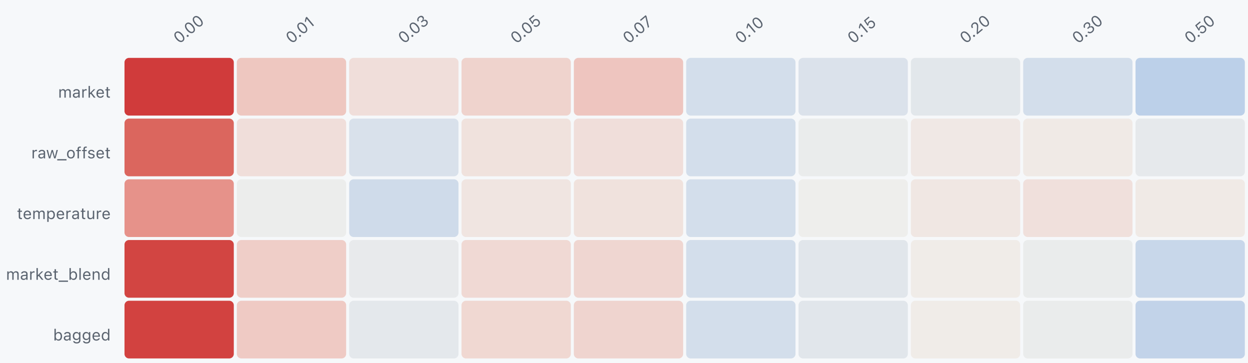
Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	68,725	291	0.0059	0.0042	0.712
0.01-0.03	90,588	1,457	0.0171	0.0161	0.941
0.03-0.05	99,693	3,713	0.0368	0.0372	1.013
0.05-0.07	73,348	4,360	0.0619	0.0594	0.960
0.07-0.10	57,417	4,731	0.0863	0.0824	0.955
0.10-0.15	61,586	7,885	0.1224	0.1280	1.046
0.15-0.20	33,460	5,910	0.1725	0.1766	1.024
0.20-0.30	32,197	7,753	0.2422	0.2408	0.994
0.30-0.50	16,224	6,042	0.3691	0.3724	1.009
0.50-1.00	2,057	1,233	0.5644	0.5994	1.062

BAGGED

The bagged probabilities, pooled over every test year.

bin	runners	wins	predicted probability	observed frequency	observed / expected
0.00-0.01	13,735	58	0.0059	0.0042	0.710
0.01-0.03	18,144	290	0.0171	0.0160	0.935
0.03-0.05	19,921	746	0.0368	0.0374	1.018
0.05-0.07	14,672	871	0.0619	0.0594	0.959
0.07-0.10	11,473	942	0.0863	0.0821	0.952
0.10-0.15	12,315	1,575	0.1223	0.1279	1.045
0.15-0.20	6,700	1,184	0.1725	0.1767	1.025
0.20-0.30	6,440	1,550	0.2422	0.2407	0.994
0.30-0.50	3,249	1,211	0.3690	0.3727	1.010
0.50-1.00	410	248	0.5646	0.6049	1.071



Observed over expected by stage and bin Observed wins over expected wins, on fixed report-wide bins, pooled over years rather than averaged across them. Blue is over-delivery, red under. Colour saturates at +0.30 from 1.0, so a cell further out than that is drawn at the endpoint and its value must be read from the tables above.

The largest departure is the 0.000 to 0.010 bin, and every stage misses there: market 0.49, raw_offset 0.76, temperature 0.84, market_blend 0.71, bagged 0.71. market is furthest from 1.0 at 0.49, meaning that bin won fewer races than it priced; temperature is closest at 0.84, so this bin is where the pipeline moves calibration most. On the market stage that bin holds 12,309 runners, the 6th smallest of 10 bins, so the departure is not a thin-bin artefact. Calibration in this bin does not improve monotonically through the pipeline: temperature reaches 0.84 and market_blend falls back to 0.71, so that step gives back part of the correction rather than adding to it.

SHARPNESS

stage	races	mean entropy	mean normalised entropy	mean top probability	mean field size
market	8,675	2.0683	0.8263	0.2913	12.3411
raw_offset	8,675	2.0296	0.8109	0.3044	12.3411
temperature	8,675	2.0181	0.8063	0.3084	12.3411
market_blend	8,675	2.0415	0.8156	0.3005	12.3411
bagged	8,675	2.0416	0.8156	0.3005	12.3411

Entropy normalised by field size, so an eight-runner race and a fourteen-runner race are on one scale, and averaged over years weighted by races rather than as a mean of yearly means. A sharper distribution is not automatically better: it is only better if the calibration above holds.

Top-choice movement

test year	n races	n disagreement races	disagreement share	model top win rate	market top win rate	disagreement mean delta	agreement mean delta	seed top choice agreement
2015	742	46	0.061995	0.239130	0.260870	-0.015097	-0.005260	0.839130
2016	744	52	0.069892	0.250000	0.250000	-0.013700	-0.004499	0.965385
2017	772	56	0.072539	0.321429	0.107143	-0.017866	-0.003062	0.871429
2018	756	60	0.079365	0.183333	0.050000	-0.001932	-0.004737	0.930000
2019	759	50	0.065876	0.300000	0.220000	-0.021397	-0.004942	0.928000
2020	799	43	0.053817	0.209302	0.186047	0.003582	-0.003787	0.902326
2021	783	70	0.089400	0.242857	0.257143	-0.017521	-0.005381	0.931429
2022	783	50	0.063857	0.240000	0.220000	-0.006338	-0.000737	0.892000
2023	796	45	0.056533	0.177778	0.222222	-0.008206	-0.003662	0.924444
2024	792	50	0.063131	0.160000	0.260000	0.004092	-0.008317	0.892000
2025	748	59	0.078877	0.203390	0.169492	-0.006463	-0.004442	0.911864
2026	201	11	0.054726	0.090909	0.363636	0.006035	0.000726	0.909091

How often this model's first pick differs from the market's, and whether it is right when it does. Disagreeing more is not better on its own -- the useful figure is the win rate *conditional* on disagreeing, because a model that disagrees often and loses is worse than one that never disagrees. The last column is how often the seeds pick the same runner as each other, which bounds how much of any disagreement is signal rather than seed noise.

Ensemble stability

RUNNERS COVERED	YEARS	SEEDS BAGGED
107,059	12	5

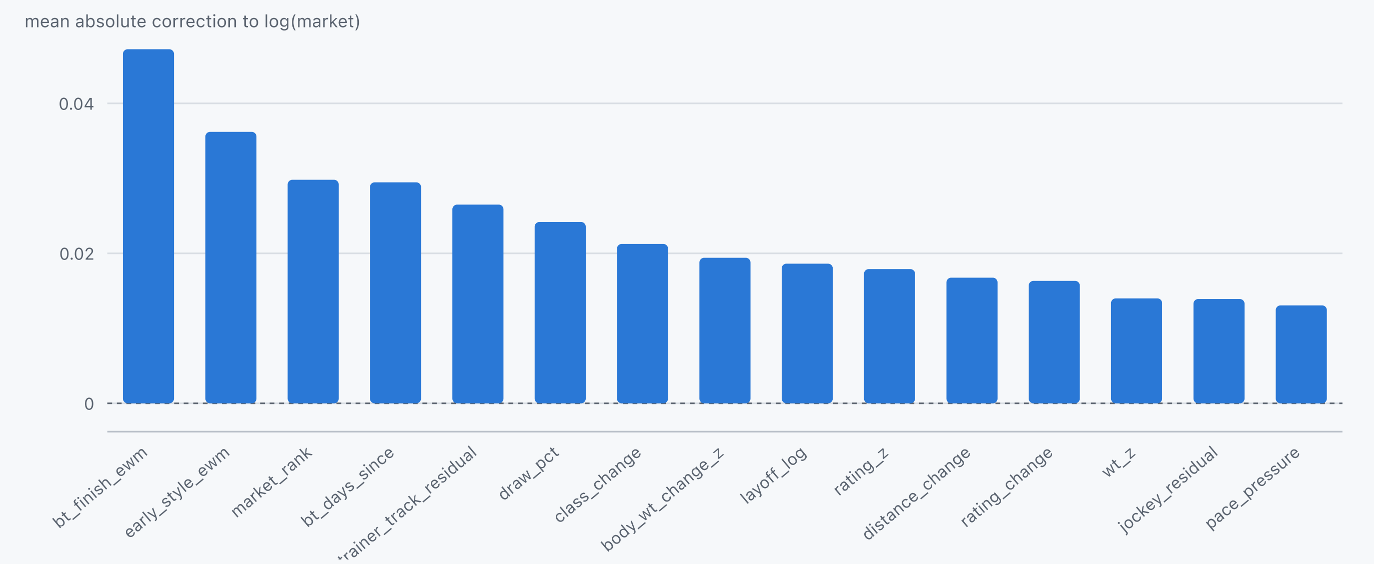
Member dispersion per runner. The bag is only worth its cost if members genuinely disagree: an ensemble whose members agree everywhere is one fit with five times the runtime. The per-runner spread across members is in `ensemble_diagnostics.parquet`, and the representative-races section below shows the races where the members disagreed most.

Feature and correction effects

> **These are corrections, not drivers of winning**
>
> The booster learns a correction to log(Implied_Prob), so a feature's attributed effect is its disagreement with the market -- not its contribution to a horse winning. Read the naive way, the sign inverts.

> **1 feature cannot be tested this way.**
>
> market_rank belongs to no group, so no ablation can remove it and this method cannot measure its held-out value.

label	feature	mean abs correction	sign agreement	rank stability	seed variation	psi	practical support	feature type	n years
stable_nonlinear	bt_finish_ewm	0.047216	0.991667	0.775991	0.058443	0.046310	12	continuous	12
stable_positive	early_style_ewm	0.036186	0.983333	0.570197	0.075225	0.007011	12	continuous	12
stable_nonlinear	market_rank	0.029815	1	0.461308	0.104674	0.001269	12	continuous	12
stable_nonlinear	bt_days_since	0.029458	0.833333	0.248826	0.100496	0.093612	12	continuous	12
stable_positive	trainer_track_residual	0.026494	0.950000	0.241158	0.080158	0.129261	12	continuous	12
stable_nonlinear	draw_pct	0.024172	0.925000	0.373624	0.120329	0.000533	12	continuous	12
stable_nonlinear	class_change	0.021257	0.766667	0.359226	0.266349	0.006694	12	categorical	12
stable_nonlinear	body_wt_change_z	0.019413	0.950000	0.238410	0.131227	0.010553	12	continuous	12
stable_nonlinear	layoff_log	0.018617	0.941667	0.322228	0.114803	0.103384	12	continuous	12
stable_nonlinear	rating_z	0.017888	0.925000	0.289509	0.249088	0.021924	12	continuous	12
stable_nonlinear	distance_change	0.016749	0.958333	0.193844	0.167273	0.019978	12	continuous	12
stable_nonlinear	rating_change	0.016323	0.950000	0.313155	0.193168	0.045097	12	continuous	12
stable_nonlinear	wt_z	0.013995	0.958333	0.272166	0.161691	0.035055	12	continuous	12
distribution_shifted	jockey_residual	0.013914	0.833333	0.227441	0.126905	0.794525	12	continuous	12
stable_nonlinear	pace_pressure	0.013054	0.885417	0.193571	0.144272	0.064975	12	continuous	12
stable_nonlinear	trip_excuse_ewm	0.012064	0.796296	0.268475	0.171276	0.058036	12	continuous	12
stable_nonlinear	setup_weight_z	0.011772	0.833333	0.446406	0.272290	0.002993	12	continuous	12
stable_nonlinear	bt_last_behind	0.010384	0.722222	0.200916	0.198664	0.026302	12	continuous	12
regime_dependent	closer_setup	0.005685	0.733333	0.775991	0.210115	0.057756	12	continuous	12
stable_nonlinear	distance_log	0.003923	1	0.616682	0.190325	0.072032	12	categorical	12
regime_dependent	gear_change	0.003599	0.583333	0.258334	0.175974	0.018293	6	binary	12
stable_nonlinear	field_size	0.002224	0.833333	0.479933	0.288189	0.106798	12	categorical	12
stable_nonlinear	race_class	0.002183	1	0.464102	0.334889	0.055002	12	categorical	12
stable_nonlinear	course_variant	0.001801	1	0.395644	0.125393	0.008740	12	categorical	12
distribution_shifted	going_ord	0.001708	0.591667	0.344006	0.451781	0.355145	12	categorical	12
stable_negative	Track_Turf	0.001099	1	0.500951	0.384960	0.004738	12	binary	12
stable_negative	Racecourse	0.001089	1	0.516146	0.108297	0.002257	12	binary	12



Correction magnitude by feature How far each feature moves the correction, largest first. Magnitude only: the direction is the grid below, and a large magnitude is not by itself evidence that the movement was right.

▼ Table view

category	mean absolute correction to log(market)
bt_finish_ewm	0.047216
early_style_ewm	0.036186
market_rank	0.029815
bt_days_since	0.029458
trainer_track_residual	0.026494
draw_pct	0.024172
class_change	0.021257
body_wt_change_z	0.019413
layoff_log	0.018617

category	mean absolute correction to log(market)
rating_z	0.017888
distance_change	0.016749
rating_change	0.016323
wt_z	0.013995
jockey_residual	0.013914
pace_pressure	0.013054

DIRECTION BY FEATURE AND YEAR

Response direction by feature and year Each cell is that year's tail effect: the last response bin minus the first, averaged over seeds. Blue is a response that rises across the feature's range and red one that falls; a row of one colour is a feature whose direction held every year. Colour extent is per grid, so this shows relative consistency rather than absolute size -- the sizes are the bar chart above.

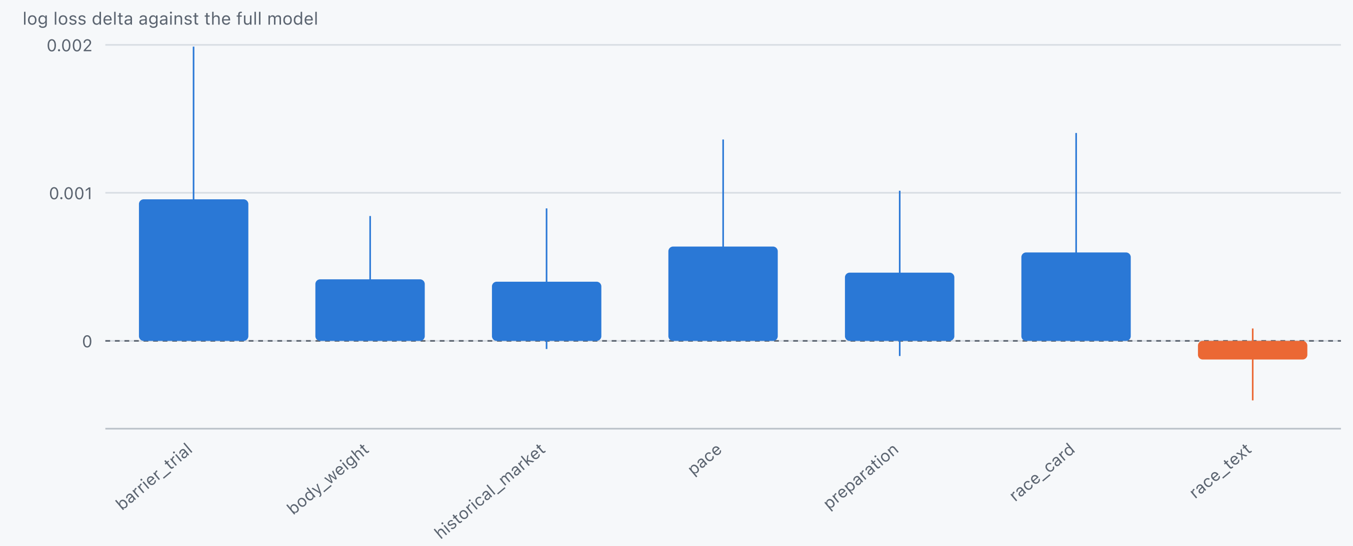
feature	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
bt_finish_ewm	0.004567	0.005879	0.005539	0.004393	0.006101	0.005372	0.006721	0.006365	0.006343	0.005667	0.006625	0.006990
early_style_ewm	0.004699	0.006102	0.006103	0.005494	0.005942	0.006982	0.008137	0.007150	0.006585	0.006344	0.006476	0.006303
market_rank	-0.002592	-0.003633	-0.003965	-0.003431	-0.002948	-0.003887	-0.004359	-0.004860	-0.004379	-0.003914	-0.004599	-0.004939
bt_days_since	-0.004123	-0.005793	-0.006845	-0.007643	-0.007718	-0.006902	-0.007143	-0.005097	-0.003971	-0.002321	-0.001852	-0.001765
trainer_track_residual	0.001226	0.003701	0.003620	0.003604	0.003463	0.004160	0.004985	0.004174	0.004264	0.004375	0.005257	0.004503
draw_pct	0.001928	0.001686	0.003571	0.002926	0.001287	0.001111	-0.000415	-0.001017	-0.001093	-0.001178	-0.000049	0.000157
class_change	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	0.006293	0.000161	0.001214	n/a
body_wt_change_z	0.000992	0.000999	0.001379	0.001441	0.001527	0.000218	0.000072	-0.000600	-0.000116	-0.000117	0.000190	0.000110
layoff_log	-0.002150	-0.001927	-0.002201	-0.001947	-0.002509	-0.002468	-0.001473	-0.001810	-0.001824	-0.001564	-0.001732	-0.001472
rating_z	-0.001628	-0.000783	0.000273	-0.000142	-0.000444	-0.001117	-0.001160	-0.001683	-0.001776	-0.001982	-0.001382	-0.000556
distance_change	-0.001252	-0.001107	-0.000867	-0.001204	-0.001946	-0.002259	-0.001937	-0.001747	-0.001920	-0.001218	-0.001026	-0.000703
rating_change	0.000765	0.000513	-0.000065	0.000030	0.000146	0.000029	-0.000600	-0.000621	-0.000353	-0.000106	-0.001267	-0.000752
wt_z	-0.000334	-0.000173	-0.000536	-0.000985	-0.001417	-0.002134	-0.001731	-0.002035	-0.001243	-0.001686	-0.001587	-0.001601
jockey_residual	0.003736	0.003379	0.002003	0.002173	0.001792	0.002275	0.003196	0.003416	0.002877	0.000741	0.000756	0.000680

feature	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
pace_pressure	-0.000959	-0.002033	-0.001709	-0.000799	0.000126	-0.000129	-0.000853	-0.000843	-0.000807	-0.000666	-0.000548	-0.000721
trip_excuse_ewm	0.001115	0.000692	0.000702	0.000317	-0.000257	-0.000350	0.000164	0.000134	-0.000178	0.000384	0.000667	0.000666
setup_weight_z	0.000163	0.001139	0.002068	0.002580	0.002121	0.002113	0.002905	0.002280	0.002098	0.000942	0.001298	0.001441
bt_last_behind	0.000780	0.001340	0.001168	0.000315	0.000320	0.000471	0.000813	0.000989	0.000844	0.000603	0.000648	0.000627
closer_setup	-0.000352	-0.000326	0.000013	-0.000052	0.000031	-0.000358	0.000025	0.000179	0.000296	0.000337	0.000422	0.000270
distance_log	0.000949	0.001779	0.000342	0.002549	0.001079	0.000567	0.002247	0.001137	0.001852	0.001370	0.004053	0.001323
gear_change	0.000382	0.000650	0.000509	0.000349	0.000321	0.000108	0.000068	-0.000070	-0.000124	-0.000120	-0.000118	-0.000184
field_size	n/a	n/a	n/a	n/a	-0.000374	n/a	n/a	n/a	n/a	n/a	n/a	n/a
race_class	0.000618	0.001006	0.000030	-0.000275	0.000077	-0.000443	-0.000724	-0.001189	-0.000933	-0.001000	-0.000867	-0.000778
course_variant	0.000972	0.000720	0.000902	0.000350	0.000418	0.000302	0.000262	-0.000075	-0.000245	-0.000036	-0.000250	-0.000242
going_ord	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	-0.002822	n/a	n/a	n/a
Track_Turf	-0.001057	-0.001087	-0.001573	-0.000829	-0.001225	-0.001056	-0.001510	-0.002079	-0.001971	-0.002186	-0.002848	-0.002545
Racecourse	-0.000438	-0.000496	-0.000439	-0.000383	-0.000242	-0.000384	-0.000419	-0.000492	-0.000762	-0.000546	-0.000703	-0.000812

15 of 27 features keep the same sign in every year they appear in. For the seven features that are constant within a race -- a racecourse, a surface -- the cell is the correction *magnitude* rather than a direction, because a race-constant feature cancels from every pairwise utility difference and has no direction to report. Which basis a row uses is in its tooltip and in the stability table's curve_basis column. A feature whose row changes colour is not necessarily broken: it may genuinely matter differently in different years. But its mean effect is then a summary of disagreeing years, and the mean is the least informative number about it.

GROUP ABLATION

group	mean delta vs full	bootstrap lower	bootstrap upper	n years	n seeds
barrier_trial	0.000956	0.000279	0.001988	12	5
body_weight	0.000415	0.000178	0.000844	12	5
historical_market	0.000400	-0.000055	0.000895	12	5
pace	0.000637	-0.000003	0.001360	12	5
preparation	0.000461	-0.000102	0.001015	12	5
race_card	0.000598	0.000150	0.001404	12	5
race_text	-0.000126	-0.000402	0.000083	12	5



Held-out value by feature group Each group removed and the model refitted, scored held out, with its bootstrap interval. Positive means removing the group made the model worse, so the group was carrying value. A bar whose interval spans zero was not distinguishable from noise.

▼ Table view

category	log loss delta against the full model	bootstrap interval
barrier_trial	0.000956	0.000279 to 0.001988
body_weight	0.000415	0.000178 to 0.000844
historical_market	0.0004	-0.000055 to 0.000895
pace	0.000637	-0.000003 to 0.00136
preparation	0.000461	-0.000102 to 0.001015
race_card	0.000598	0.00015 to 0.001404

category	log loss delta against the full model	bootstrap interval
race_text	-0.000126	-0.000402 to 0.000083

The deltas do *not* sum to the whole: features share information, so attributing a total by adding these up overstates it.

Representative races

Cohorts chosen deterministically from the artifact rather than picked by hand. A hand-picked race is an illustration; a cohort-selected one is evidence. Each cohort shows one race, with every runner's probability at each stage of the pipeline.

LARGEST ABSOLUTE MARKET DISAGREEMENT

Race `2020122604`, 2020, 14 runners.

horse number	market prob	raw corrected prob	temperature prob	bagged prob	member min	member max	member std	estimated edge	actual result
1	0.009451	0.005767	0.005313	0.006772	0.006487	0.007008	0.000197	n/a	lost
2	0.026524	0.019456	0.018420	0.021010	0.020487	0.021335	0.000341	n/a	lost
3	0.074749	0.059826	0.058088	0.063366	0.062290	0.065016	0.001113	n/a	lost
4	0.010678	0.005492	0.005054	0.006983	0.006765	0.007060	0.000123	n/a	lost
5	0.006374	0.005085	0.004672	0.005203	0.005050	0.005337	0.000109	n/a	lost
6	0.003329	0.002294	0.002070	0.002499	0.002436	0.002551	0.000047	n/a	lost
7	0.124582	0.107552	0.105811	0.114270	0.112366	0.116594	0.002089	n/a	lost
8	0.045680	0.031262	0.029914	0.035285	0.034836	0.035846	0.000380	n/a	lost
9	0.456799	0.591810	0.604948	0.552438	0.546019	0.558495	0.004676	-0.005611	won
10	0.048367	0.032048	0.030684	0.036719	0.036043	0.037292	0.000449	n/a	lost
11	0.120917	0.089801	0.087990	0.099548	0.098822	0.100531	0.000621	n/a	lost
12	0.029366	0.023954	0.022785	0.024846	0.024517	0.025183	0.000278	n/a	lost
13	0.004031	0.002815	0.002552	0.003047	0.002970	0.003102	0.000054	n/a	lost
14	0.039154	0.022838	0.021700	0.028013	0.027403	0.028731	0.000548	n/a	lost

MEDIAN MARKET DISAGREEMENT

Race `2018040801`, 2018, 14 runners.

horse number	market prob	raw corrected prob	temperature prob	bagged prob	member min	member max	member std	estimated edge	actual result
1	0.011930	0.009787	0.009565	0.010442	0.010261	0.010736	0.000185	n/a	lost
2	0.042696	0.037399	0.037007	0.038480	0.037644	0.039825	0.000845	n/a	lost
3	0.057945	0.054981	0.054598	0.054733	0.053195	0.056256	0.001184	n/a	lost
4	0.073748	0.062108	0.061746	0.066249	0.065487	0.067691	0.000906	n/a	lost
5	0.018437	0.017043	0.016743	0.017303	0.016868	0.017632	0.000359	n/a	lost
6	0.067602	0.070147	0.069816	0.068486	0.067355	0.070001	0.001027	n/a	lost
7	0.193150	0.217565	0.218813	0.210357	0.206102	0.214286	0.003348	n/a	won
8	0.253509	0.258221	0.260113	0.258901	0.256060	0.263651	0.002990	n/a	lost
9	0.082779	0.076056	0.075754	0.080244	0.079066	0.081431	0.001085	n/a	lost
10	0.021925	0.018129	0.017819	0.019678	0.019340	0.020155	0.000328	n/a	lost
11	0.005167	0.003249	0.003143	0.004066	0.004013	0.004146	0.000050	n/a	lost
12	0.036874	0.035090	0.034702	0.034833	0.034206	0.035778	0.000582	n/a	lost
13	0.124805	0.132394	0.132544	0.127876	0.126514	0.129880	0.001411	n/a	lost
14	0.009433	0.007830	0.007636	0.008351	0.008216	0.008526	0.000126	n/a	lost

NEAR ZERO MARKET DISAGREEMENT

Race `2015010102`, 2015, 7 runners.

horse number	market prob	raw corrected prob	temperature prob	bagged prob	member min	member max	member std	estimated edge	actual result
1	0.073896	0.076165	0.075237	0.074802	0.073220	0.077638	0.001676	n/a	lost
2	0.073896	0.077734	0.076808	0.073347	0.071862	0.075118	0.001314	n/a	lost
3	0.086474	0.093058	0.092175	0.088951	0.087428	0.090608	0.001203	n/a	lost
4	0.225793	0.222524	0.223039	0.224341	0.223184	0.226068	0.001111	n/a	won
5	0.088354	0.097784	0.096922	0.091239	0.090326	0.092606	0.001012	n/a	lost
6	0.045159	0.041376	0.040534	0.043006	0.042392	0.043394	0.000378	n/a	lost

horse number	market prob	raw corrected prob	temperature prob	bagged prob	member min	member max	member std	estimated edge	actual result
8	0.406428	0.391359	0.395285	0.404314	0.400962	0.410066	0.003596	n/a	lost

LARGEST MEMBER DISAGREEMENT

Race `2018040809`, 2018, 14 runners.

horse number	market prob	raw corrected prob	temperature prob	bagged prob	member min	member max	member std	estimated edge	actual result
1	0.085019	0.072355	0.071790	0.079738	0.078343	0.081416	0.001106	n/a	lost
2	0.010739	0.010026	0.009768	0.010319	0.010205	0.010533	0.000130	n/a	lost
3	0.029149	0.030508	0.030030	0.029715	0.029300	0.030528	0.000475	n/a	lost
4	0.272062	0.310587	0.312337	0.303324	0.292388	0.310721	0.007708	n/a	lost
5	0.022059	0.016174	0.015827	0.018655	0.018421	0.018914	0.000201	n/a	won
6	0.011829	0.009327	0.009081	0.010439	0.010171	0.010620	0.000165	n/a	lost
7	0.354863	0.367179	0.369819	0.349041	0.336882	0.362411	0.010483	n/a	lost
8	0.003962	0.002968	0.002859	0.003396	0.003360	0.003448	0.000035	n/a	lost
9	0.074199	0.068215	0.067646	0.071484	0.070032	0.073022	0.001240	n/a	lost
10	0.024733	0.017719	0.017354	0.020747	0.020477	0.021009	0.000255	n/a	lost
11	0.005007	0.003306	0.003188	0.004051	0.003994	0.004089	0.000038	n/a	lost
12	0.034008	0.031472	0.030987	0.032709	0.032180	0.033211	0.000403	n/a	lost
13	0.058299	0.047780	0.047226	0.053267	0.051793	0.054551	0.001071	n/a	lost
14	0.014072	0.012384	0.012088	0.013113	0.012751	0.013368	0.000235	n/a	lost

LARGEST CORRECT CORRECTION

Race `2017110504`, 2017, 12 runners.

horse number	market prob	raw corrected prob	temperature prob	bagged prob	member min	member max	member std	estimated edge	actual result
1	0.074039	0.049241	0.047762	0.060124	0.059456	0.060840	0.000626	n/a	lost
2	0.037019	0.022997	0.022019	0.029146	0.028736	0.029759	0.000401	n/a	lost
3	0.110058	0.062337	0.060708	0.084594	0.083372	0.086696	0.001246	n/a	lost
4	0.019864	0.012757	0.012093	0.015641	0.015388	0.016045	0.000265	n/a	lost
5	0.542952	0.687631	0.697607	0.622930	0.617359	0.626581	0.003445	n/a	won
6	0.018510	0.011425	0.010810	0.014199	0.013875	0.014578	0.000296	n/a	lost
7	0.107162	0.095625	0.093806	0.101747	0.099940	0.104834	0.001904	n/a	lost
8	0.012927	0.009687	0.009140	0.010816	0.010541	0.011001	0.000194	n/a	lost
9	0.008950	0.006101	0.005711	0.007118	0.006914	0.007263	0.000140	n/a	lost
10	0.010859	0.008178	0.007694	0.009178	0.008981	0.009515	0.000210	n/a	lost
11	0.003365	0.002395	0.002207	0.002706	0.002624	0.002773	0.000056	n/a	lost
12	0.054295	0.031624	0.030444	0.041801	0.041229	0.042602	0.000586	n/a	lost

LARGEST INCORRECT CORRECTION

Race `2019022008`, 2019, 12 runners.

horse number	market prob	raw corrected prob	temperature prob	bagged prob	member min	member max	member std	estimated edge	actual result
1	0.067221	0.072843	0.072361	0.069313	0.067374	0.070584	0.001381	n/a	lost
2	0.016133	0.017360	0.016907	0.016708	0.016510	0.017090	0.000225	n/a	lost
3	0.126040	0.147714	0.148176	0.139155	0.138494	0.140268	0.000680	n/a	lost
4	0.057618	0.050900	0.050312	0.053507	0.052221	0.054724	0.000916	n/a	lost
5	0.021228	0.031464	0.030895	0.027057	0.025898	0.027900	0.000798	0.028157	lost
6	0.067221	0.077425	0.076977	0.074051	0.072466	0.075588	0.001302	n/a	lost
7	0.035072	0.044793	0.044198	0.040907	0.040384	0.041585	0.000545	n/a	lost
8	0.073333	0.059363	0.058803	0.064814	0.063924	0.066330	0.000910	n/a	lost
9	0.366663	0.294754	0.298513	0.323584	0.319497	0.327765	0.003903	n/a	won
10	0.026021	0.024469	0.023943	0.025370	0.024802	0.025623	0.000324	n/a	lost
11	0.122221	0.152453	0.152996	0.142242	0.139446	0.144776	0.002270	n/a	lost
12	0.021228	0.026461	0.025920	0.023292	0.022920	0.023980	0.000411	n/a	lost

SELECTED BETS

Race 2018062701, 2018, 12 runners.

horse number	market prob	raw corrected prob	temperature prob	bagged prob	member min	member max	member std	estimated edge	actual result
1	0.068158	0.049965	0.049587	0.059000	0.057533	0.059954	0.000924	n/a	lost
2	0.015432	0.013488	0.013225	0.014201	0.013886	0.014571	0.000293	n/a	lost
3	0.272631	0.265448	0.267536	0.269955	0.266133	0.271703	0.002232	n/a	won
4	0.148708	0.161397	0.161921	0.153768	0.149853	0.155852	0.002606	n/a	lost
5	0.022105	0.039990	0.039606	0.031385	0.030938	0.032303	0.000548	0.161261	lost
6	0.043047	0.040466	0.040081	0.042176	0.041357	0.043551	0.000880	n/a	lost
7	0.062915	0.054896	0.054529	0.059280	0.058682	0.060054	0.000583	n/a	lost
8	0.122074	0.117780	0.117819	0.118775	0.117597	0.120000	0.001126	n/a	lost
9	0.048111	0.048587	0.048208	0.048109	0.047859	0.048439	0.000219	n/a	lost
10	0.048111	0.047751	0.047370	0.047576	0.047026	0.048087	0.000402	n/a	lost
11	0.123923	0.136251	0.136479	0.131607	0.130260	0.132928	0.001217	n/a	lost
12	0.024785	0.023981	0.023639	0.024168	0.024013	0.024333	0.000120	n/a	lost

Betting and bankroll analysis

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

SETTLEMENT BY STAKING RULE

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
flat_stake@edge>=0.02	174	1,740,000	-973,500	-0.559483	-0.102420	0.098882	9,026,500	169
kelly_no_rebate@0.01	338	110,870	-578	-0.005213	-0.000058	0.002609	9,999,422	324
kelly_no_rebate@0.02	341	221,750	-947.500000	-0.004273	-0.000095	0.005229	9999052.500000	327
kelly_no_rebate@0.05	346	554,210	-2545	-0.004592	-0.000255	0.013015	9,997,455	332
kelly_no_rebate@0.1	346	1,108,400	-6718	-0.006061	-0.000672	0.025854	9,993,282	332
kelly_settled_rebate@0.01	338	110,870	-578	-0.005213	-0.000058	0.002609	9,999,422	324
kelly_settled_rebate@0.02	341	221,750	-947.500000	-0.004273	-0.000095	0.005229	9999052.500000	327
kelly_settled_rebate@0.05	346	554,210	-2545	-0.004592	-0.000255	0.013015	9,997,455	332
kelly_settled_rebate@0.1	346	1,109,420	-6444.500000	-0.005809	0.001113	0.025355	10011131.500000	332
rebate_aware_kelly@0.01	339	155,100	-10412.500000	-0.067134	-0.000742	0.004281	9992587.500000	325
rebate_aware_kelly@0.02	371	1,083,220	50671.500000	0.046779	0.011842	0.016956	10119124.500000	348
rebate_aware_kelly@0.05	772	7,045,400	-742,039	-0.105322	-0.017787	0.057885	9,823,706	728
rebate_aware_kelly@0.1	1398	17,945,530	-1,174,968	-0.065474	0.028644	0.085645	10,290,581	1302

Model view bagged_offset. Seed -1. Every rule is settled on the same selected bets at the same odds, so the differences between these rows are stake sizing alone.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

FLAT STAKE

bucket	mean estimated edge	expected return	realised at edge>=0.02	bets at edge>=0.02
0.020000 to 0.050000	0.031990	0.031990	-0.6870	115
0.050000 to 0.100000	0.069152	0.069152	-0.5160	50
0.100000 to 0.200000	0.133063	0.133063	0.8278	9

KELLY, NO REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009823	0.009823	0.0105	0.0161	0.0159	0.0155	164	167	172	172
0.020000 to 0.050000	0.031990	0.031990	-0.4455	-0.4476	-0.4485	-0.4513	115	115	115	115
0.050000 to 0.100000	0.069152	0.069152	-0.0831	-0.0821	-0.0834	-0.0863	50	50	50	50
0.100000 to 0.200000	0.133063	0.133063	3.6067	3.6013	3.6146	3.6301	9	9	9	9

KELLY, SETTLED REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009823	0.009823	0.0105	0.0161	0.0159	0.0155	164	167	172	172
0.020000 to 0.050000	0.031990	0.031990	-0.4455	-0.4476	-0.4485	-0.4512	115	115	115	115
0.050000 to 0.100000	0.069152	0.069152	-0.0831	-0.0821	-0.0834	-0.0859	50	50	50	50
0.100000 to 0.200000	0.133063	0.133063	3.6067	3.6013	3.6146	3.6321	9	9	9	9

REBATE-AWARE KELLY

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
unbounded to 0	-0.005611	-0.005611	0.8000	0.1869	-0.0671	-0.0220	1	30	427	1053
0 to 0.020000	0.009823	0.009823	-0.5201	-0.2768	-0.1540	-0.1224	164	167	171	171
0.020000 to 0.050000	0.031990	0.031990	-0.4459	-0.2440	-0.4412	-0.4780	115	115	115	115
0.050000 to 0.100000	0.069152	0.069152	0.1596	0.1275	-0.1227	-0.1348	50	50	50	50
0.100000 to 0.200000	0.133063	0.133063	3.6067	5.4892	4.0988	3.6541	9	9	9	9

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

A negative-edge bucket appears only under rebate-aware sizing, and only at the larger fractions. That is the rule working as defined rather than a filter failing: the rebate is a cliff, so a runner whose own edge is negative can pay for itself by pushing the race's turnover over the threshold, and the stakes only get large enough for that at the higher fractions.

KELLY PRESETS

OPENING BANKROLL	PRESETS SETTLED	PATH MODES	SEEDS	TRAINED ON
HK$10,000,000	1% Kelly, 2% Kelly, 5% Kelly, 10% Kelly	annual reset, continuous	5 7, 17, 42, 123, 256	2010-2014 then expanding a separate fit per test year, each on 2010 through the year before it -- 12 fits in all

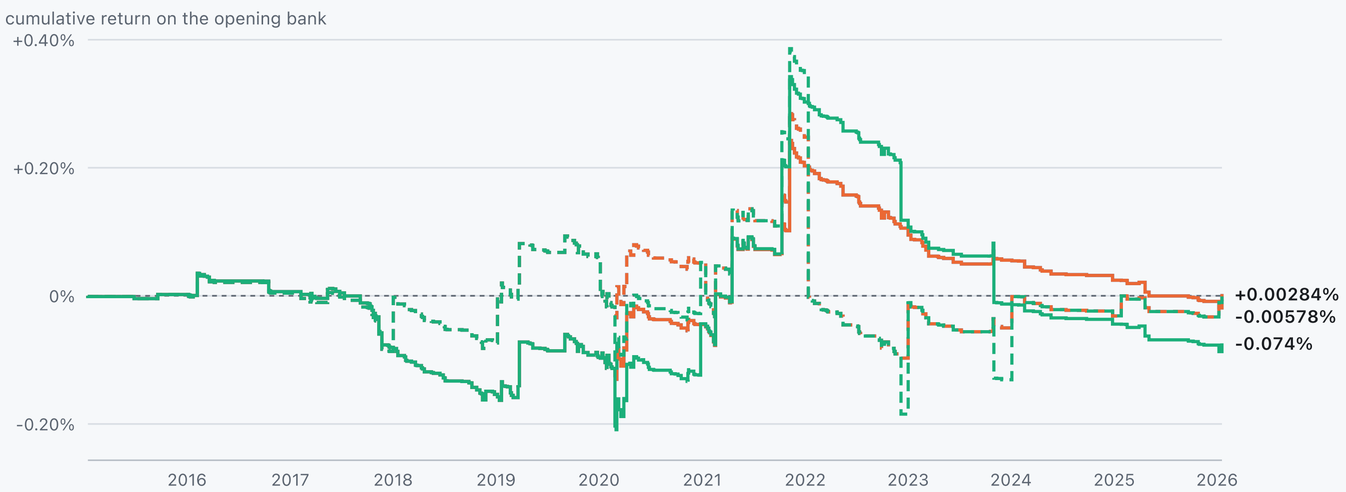
BOOSTING ROUNDS	YEARS STAKED
400	2015-2026

These are walk-forward figures, not one model's: for each test year the model was fitted on 2010 through the previous year and then scored that year, so the result concatenates 12 separately fitted models and no test year appears in its own training data. The window expands each year, so the last year's model saw the most data and the first year's the least. Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. Rebate policy: 10% of the race net loss, once that loss reaches HK$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above.

1% KELLY

OPENING BANKROLL	BEST FINAL RETURN	BETS	REBATE RECEIVED
HK$10,000,000	+0.00284% Kelly, no rebate, annual reset	338 settled bets	HK$3,000 the most any rule received

Kelly, no rebate (= Kelly, settled rebate)   Rebate-aware Kelly



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	331	+0.00284%	338 bets
Kelly, no rebate, continuous	331	-0.00578%	338 bets
Kelly, settled rebate, annual reset	331	+0.00284%	338 bets
Kelly, settled rebate, continuous	331	-0.00578%	338 bets
Rebate-aware Kelly, annual reset	332	+0.00284%	339 bets
Rebate-aware Kelly, continuous	332	-0.074%	339 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2015	3	3	3
2016	14	14	14
2017	37	37	37
2018	42	42	42
2019	49	49	49
2020	58	58	59
2021	54	54	54
2022	37	37	37
2023	21	21	21
2024	10	10	10
2025	11	11	11
2026	*2*	*2*	*2*

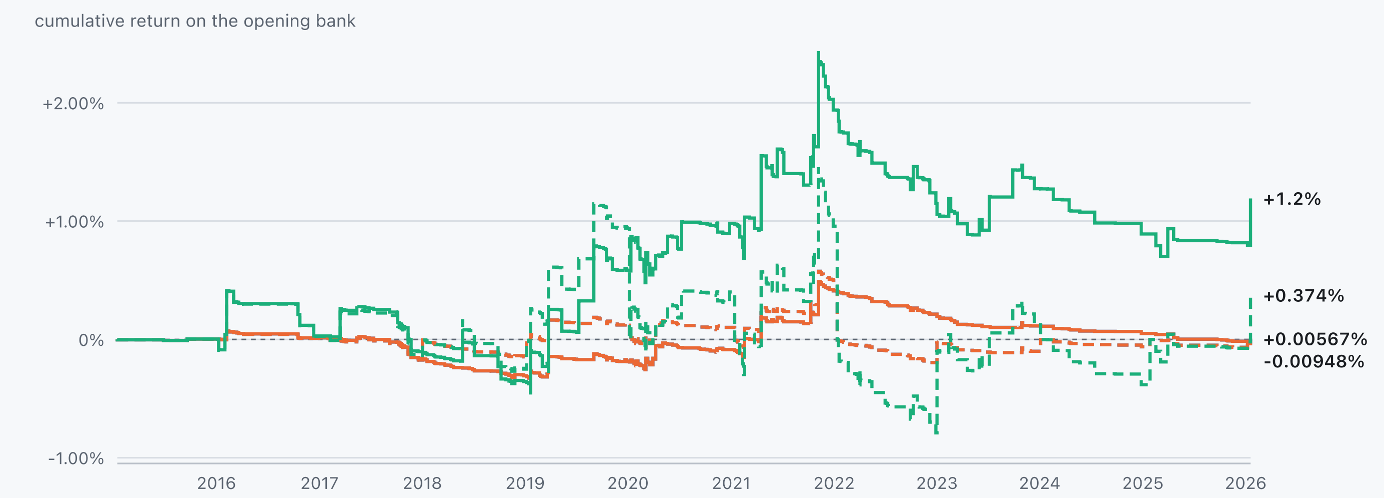
2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate and Kelly, settled rebate: 3 bets in 2015 against 11 in 2025. Rebate-aware Kelly: 3 bets in 2015 against 11 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Kelly, no rebate and Kelly, settled rebate (annual reset) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Kelly, no rebate and Kelly, settled rebate (continuous) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Rebate received: Rebate-aware Kelly, annual reset HK$3,000; Rebate-aware Kelly, continuous HK$3,000. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

2% KELLY

OPENING BANKROLL	BEST FINAL RETURN	BETS	REBATE RECEIVED
HK$10,000,000	+1.2%	371	HK$68,453
	Rebate-aware Kelly, continuous	settled bets	the most any rule received

Kelly, no rebate (= Kelly, settled rebate) Rebate-aware Kelly



Cumulative bankroll return at 2% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	334	+0.00567%	341 bets
Kelly, no rebate, continuous	334	-0.00948%	341 bets
Kelly, settled rebate, annual reset	334	+0.00567%	341 bets

Series	Settlements	Final return	Note
Kelly, settled rebate, continuous	334	-0.00948%	341 bets
Rebate-aware Kelly, annual reset	363	+0.374%	371 bets
Rebate-aware Kelly, continuous	363	+1.2%	371 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2015	3	3	3
2016	15	15	17
2017	37	37	37
2018	42	42	45
2019	50	50	53
2020	58	58	62
2021	54	54	59
2022	37	37	41
2023	22	22	28
2024	10	10	11
2025	11	11	13
2026	*2*	*2*	*2*

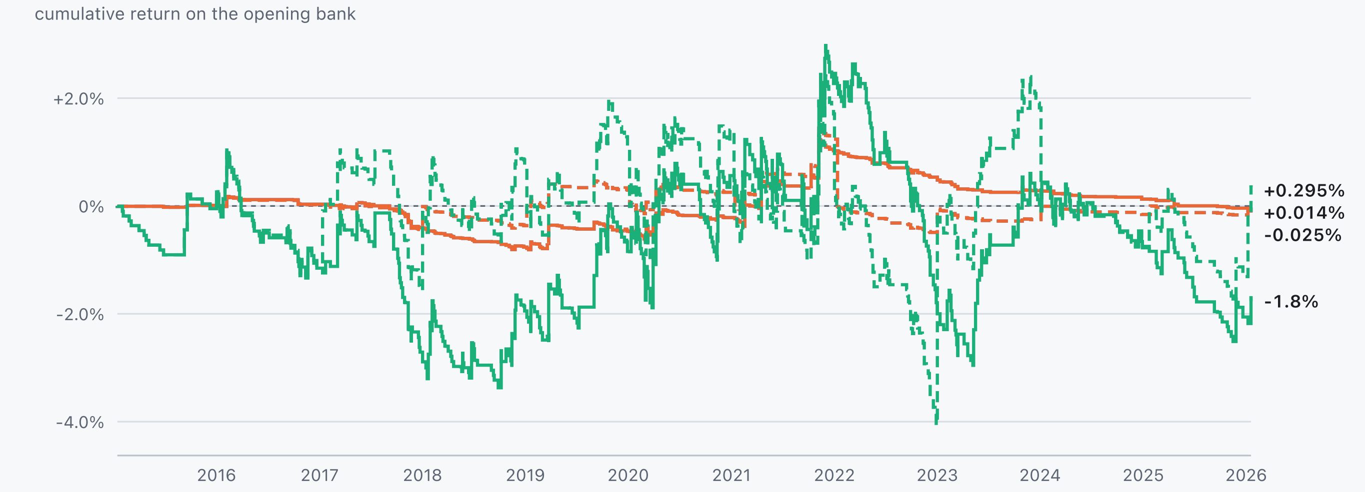
2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate and Kelly, settled rebate: 3 bets in 2015 against 11 in 2025. Rebate-aware Kelly: 3 bets in 2015 against 13 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Kelly, no rebate and Kelly, settled rebate (annual reset) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Kelly, no rebate and Kelly, settled rebate (continuous) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Rebate received: Rebate-aware Kelly, annual reset HK$68,453; Rebate-aware Kelly, continuous HK$68,453. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

5% KELLY

OPENING BANKROLL	BEST FINAL RETURN	BETS	REBATE RECEIVED
HK$10,000,000	+0.295%	772	HK$565,745
	Rebate-aware Kelly, annual reset	settled bets	the most any rule received

Kelly, no rebate (= Kelly, settled rebate) Rebate-aware Kelly



Cumulative bankroll return at 5% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	339	+0.014%	346 bets
Kelly, no rebate, continuous	339	-0.025%	346 bets
Kelly, settled rebate, annual reset	339	+0.014%	346 bets
Kelly, settled rebate, continuous	339	-0.025%	346 bets
Rebate-aware Kelly, annual reset	740	+0.295%	772 bets
Rebate-aware Kelly, continuous	740	-1.8%	772 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2015	3	3	16
2016	16	16	50
2017	39	39	72
2018	42	42	80
2019	50	50	77
2020	58	58	110
2021	54	54	124
2022	39	39	98
2023	22	22	61
2024	10	10	44
2025	11	11	37
2026	*2*	*2*	*3*

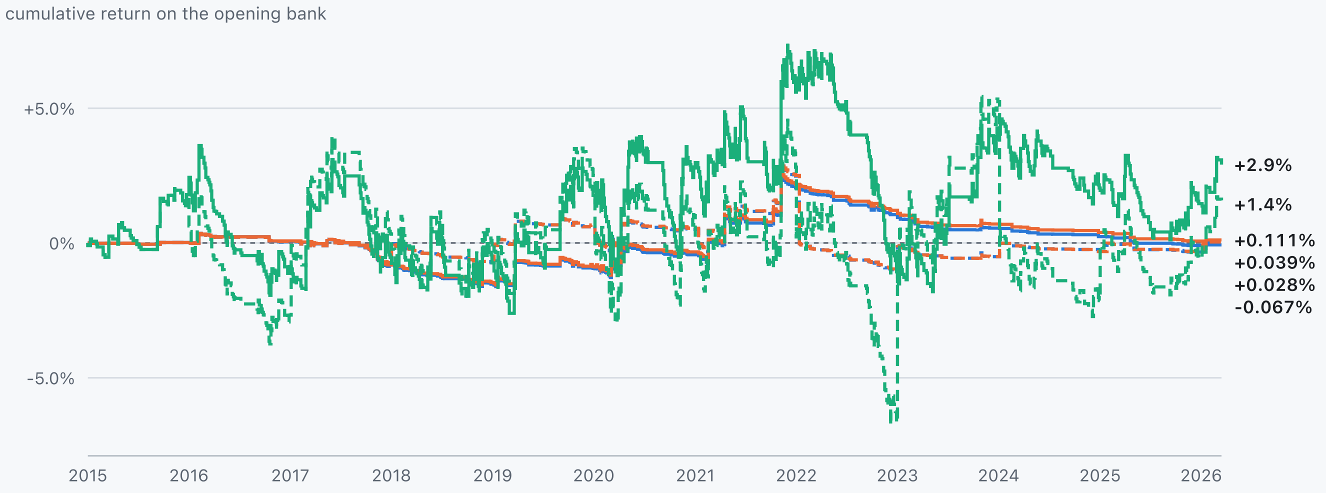
2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate and Kelly, settled rebate: 3 bets in 2015 against 11 in 2025. Rebate-aware Kelly: 16 bets in 2015 against 37 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Kelly, no rebate and Kelly, settled rebate (annual reset) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Kelly, no rebate and Kelly, settled rebate (continuous) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Rebate received: Rebate-aware Kelly, annual reset HK$565,745; Rebate-aware Kelly, continuous HK$565,745. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

10% KELLY

OPENING BANKROLL	BEST FINAL RETURN	BETS	REBATE RECEIVED
HK$10,000,000	+2.9%	1,398	HK$1,465,549
	Rebate-aware Kelly, continuous	settled bets	the most any rule received

Kelly, no rebate · Kelly, settled rebate · Rebate-aware Kelly



Cumulative bankroll return at 10% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	339	+0.028%	346 bets
Kelly, no rebate, continuous	339	-0.067%	346 bets
Kelly, settled rebate, annual reset	339	+0.039%	346 bets
Kelly, settled rebate, continuous	339	+0.111%	346 bets
Rebate-aware Kelly, annual reset	1,323	+1.4%	1,398 bets
Rebate-aware Kelly, continuous	1,323	+2.9%	1,398 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2015	3	3	44
2016	16	16	105
2017	39	39	121
2018	42	42	138
2019	50	50	117
2020	58	58	188
2021	54	54	198
2022	39	39	186
2023	22	22	112
2024	10	10	87
2025	11	11	86
2026	*2*	*2*	*16*

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate and Kelly, settled rebate: 3 bets in 2015 against 11 in 2025. Rebate-aware Kelly: 44 bets in 2015 against 86 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK$1,465,549; Rebate-aware Kelly, continuous HK$1,465,549; Kelly, settled rebate, annual reset HK$17,576; Kelly, settled rebate, continuous HK$17,576. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK$10,000,000	HK$10,000,284	+0.00284%	338	HK$0
1% Kelly	Kelly, no rebate, continuous	HK$10,000,000	HK$9,999,422	-0.00578%	338	HK$0
1% Kelly	Kelly, settled rebate, annual reset	HK$10,000,000	HK$10,000,284	+0.00284%	338	HK$0
1% Kelly	Kelly, settled rebate, continuous	HK$10,000,000	HK$9,999,422	-0.00578%	338	HK$0

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Rebate-aware Kelly, annual reset	HK$10,000,000	HK$10,000,284	+0.00284%	339	HK$3,000
1% Kelly	Rebate-aware Kelly, continuous	HK$10,000,000	HK$9,992,588	-0.074%	339	HK$3,000
2% Kelly	Kelly, no rebate, annual reset	HK$10,000,000	HK$10,000,567	+0.00567%	341	HK$0
2% Kelly	Kelly, no rebate, continuous	HK$10,000,000	HK$9,999,052	-0.00948%	341	HK$0
2% Kelly	Kelly, settled rebate, annual reset	HK$10,000,000	HK$10,000,567	+0.00567%	341	HK$0
2% Kelly	Kelly, settled rebate, continuous	HK$10,000,000	HK$9,999,052	-0.00948%	341	HK$0
2% Kelly	Rebate-aware Kelly, annual reset	HK$10,000,000	HK$10,037,440	+0.374%	371	HK$68,453
2% Kelly	Rebate-aware Kelly, continuous	HK$10,000,000	HK$10,119,124	+1.2%	371	HK$68,453
5% Kelly	Kelly, no rebate, annual reset	HK$10,000,000	HK$10,001,408	+0.014%	346	HK$0
5% Kelly	Kelly, no rebate, continuous	HK$10,000,000	HK$9,997,455	-0.025%	346	HK$0
5% Kelly	Kelly, settled rebate, annual reset	HK$10,000,000	HK$10,001,408	+0.014%	346	HK$0
5% Kelly	Kelly, settled rebate, continuous	HK$10,000,000	HK$9,997,455	-0.025%	346	HK$0
5% Kelly	Rebate-aware Kelly, annual reset	HK$10,000,000	HK$10,029,518	+0.295%	772	HK$565,745
5% Kelly	Rebate-aware Kelly, continuous	HK$10,000,000	HK$9,823,706	-1.8%	772	HK$565,745
10% Kelly	Kelly, no rebate, annual reset	HK$10,000,000	HK$10,002,815	+0.028%	346	HK$0
10% Kelly	Kelly, no rebate, continuous	HK$10,000,000	HK$9,993,282	-0.067%	346	HK$0
10% Kelly	Kelly, settled rebate, annual reset	HK$10,000,000	HK$10,003,868	+0.039%	346	HK$17,576
10% Kelly	Kelly, settled rebate, continuous	HK$10,000,000	HK$10,011,132	+0.111%	346	HK$17,576
10% Kelly	Rebate-aware Kelly, annual reset	HK$10,000,000	HK$10,144,542	+1.4%	1,398	HK$1,465,549
10% Kelly	Rebate-aware Kelly, continuous	HK$10,000,000	HK$10,290,581	+2.9%	1,398	HK$1,465,549

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

DAY-BLOCK BOOTSTRAP OF THE BANKROLL

series	replicates	median terminal bankroll	terminal bankroll p5	p below initial	median max drawdown	p95 max drawdown	p below min bet
flat_stake@edge>=0.02 / block1	2000	HK$9,011,645	HK$8,701,055	1.0000	0.1001	0.1304	0.0000
flat_stake@edge>=0.02 / block3	2000	HK$9,027,154	HK$8,722,326	1.0000	0.0988	0.1292	0.0000
flat_stake@edge>=0.02 / block7	2000	HK$9,022,940	HK$8,676,218	0.9995	0.0992	0.1339	0.0000
kelly_no_rebate@0.01 / block1	2000	HK$9,998,828	HK$9,962,064	0.5175	0.0027	0.0042	0.0000
kelly_no_rebate@0.01 / block3	2000	HK$9,997,755	HK$9,965,458	0.5385	0.0027	0.0039	0.0000
kelly_no_rebate@0.01 / block7	2000	HK$9,998,701	HK$9,968,259	0.5265	0.0025	0.0037	0.0000
kelly_no_rebate@0.02 / block1	2000	HK$9,997,322	HK$9,925,704	0.5230	0.0054	0.0083	0.0000
kelly_no_rebate@0.02 / block3	2000	HK$9,997,433	HK$9,937,364	0.5220	0.0052	0.0073	0.0000
kelly_no_rebate@0.02 / block7	2000	HK$9,999,750	HK$9,948,096	0.5035	0.0051	0.0067	0.0000
kelly_no_rebate@0.05 / block1	2000	HK$9,992,713	HK$9,818,800	0.5275	0.0135	0.0205	0.0000
kelly_no_rebate@0.05 / block3	2000	HK$9,997,740	HK$9,838,260	0.5055	0.0131	0.0187	0.0000
kelly_no_rebate@0.05 / block7	2000	HK$9,996,284	HK$9,828,915	0.5145	0.0125	0.0201	0.0000
kelly_no_rebate@0.1 / block1	2000	HK$9,983,809	HK$9,639,911	0.5290	0.0269	0.0406	0.0000
kelly_no_rebate@0.1 / block3	2000	HK$9,993,867	HK$9,678,171	0.5075	0.0260	0.0371	0.0000
kelly_no_rebate@0.1 / block7	2000	HK$9,990,857	HK$9,659,398	0.5175	0.0249	0.0399	0.0000
kelly_settled_rebate@0.01 / block1	2000	HK$9,998,828	HK$9,962,064	0.5175	0.0027	0.0042	0.0000
kelly_settled_rebate@0.01 / block3	2000	HK$9,997,755	HK$9,965,458	0.5385	0.0027	0.0039	0.0000
kelly_settled_rebate@0.01 / block7	2000	HK$9,998,701	HK$9,968,259	0.5265	0.0025	0.0037	0.0000
kelly_settled_rebate@0.02 / block1	2000	HK$9,997,322	HK$9,925,704	0.5230	0.0054	0.0083	0.0000
kelly_settled_rebate@0.02 / block3	2000	HK$9,997,433	HK$9,937,364	0.5220	0.0052	0.0073	0.0000
kelly_settled_rebate@0.02 / block7	2000	HK$9,999,750	HK$9,948,096	0.5035	0.0051	0.0067	0.0000
kelly_settled_rebate@0.05 / block1	2000	HK$9,992,713	HK$9,818,800	0.5275	0.0135	0.0205	0.0000

series	replicates	median terminal bankroll	terminal bankroll p5	p below initial	median max drawdown	p95 max drawdown	p below min bet
kelly_settled_rebate@0.05 / block3	2000	HK$9,997,740	HK$9,838,260	0.5055	0.0131	0.0187	0.0000
kelly_settled_rebate@0.05 / block7	2000	HK$9,996,284	HK$9,828,915	0.5145	0.0125	0.0201	0.0000
kelly_settled_rebate@0.1 / block1	2000	HK$10,001,580	HK$9,656,484	0.4975	0.0262	0.0388	0.0000
kelly_settled_rebate@0.1 / block3	2000	HK$10,012,580	HK$9,697,626	0.4775	0.0253	0.0356	0.0000
kelly_settled_rebate@0.1 / block7	2000	HK$10,007,943	HK$9,677,738	0.4845	0.0242	0.0384	0.0000
rebate_aware_kelly@0.01 / block1	2000	HK$9,992,109	HK$9,940,904	0.5870	0.0042	0.0068	0.0000
rebate_aware_kelly@0.01 / block3	2000	HK$9,993,762	HK$9,947,417	0.5905	0.0040	0.0065	0.0000
rebate_aware_kelly@0.01 / block7	2000	HK$9,992,879	HK$9,952,480	0.6110	0.0041	0.0061	0.0000
rebate_aware_kelly@0.02 / block1	2000	HK$10,126,283	HK$9,846,592	0.2430	0.0171	0.0282	0.0000
rebate_aware_kelly@0.02 / block3	2000	HK$10,123,226	HK$9,869,781	0.2250	0.0164	0.0263	0.0000
rebate_aware_kelly@0.02 / block7	2000	HK$10,114,399	HK$9,867,028	0.2225	0.0147	0.0250	0.0000
rebate_aware_kelly@0.05 / block1	2000	HK$9,816,230	HK$8,973,151	0.6355	0.0728	0.1206	0.0000
rebate_aware_kelly@0.05 / block3	2000	HK$9,806,013	HK$8,937,516	0.6405	0.0752	0.1240	0.0000
rebate_aware_kelly@0.05 / block7	2000	HK$9,843,809	HK$8,985,885	0.6000	0.0710	0.1187	0.0000
rebate_aware_kelly@0.1 / block1	2000	HK$10,255,008	HK$8,553,161	0.4205	0.1223	0.2099	0.0000
rebate_aware_kelly@0.1 / block3	2000	HK$10,241,534	HK$8,504,669	0.4165	0.1239	0.2176	0.0000
rebate_aware_kelly@0.1 / block7	2000	HK$10,298,927	HK$8,621,977	0.3920	0.1203	0.2038	0.0000

Race days resampled with replacement, so results from one meeting stay together. This is sampling risk in the settled path, not model risk. 39 of 234 settled series shown, filtered to model view `bagged_offset`.

MODEL-ASSUMED MONTE CARLO

series	replicates	median terminal bankroll	terminal bankroll p5	p below initial	median max drawdown	p95 max drawdown	p below min bet
flat_stake@edge>=0.02	1000	HK$10,001,943	HK$9,203,076	0.4980	0.0514	0.0920	0.0000
kelly_no_rebate@0.01	1000	HK$10,003,246	HK$9,969,687	0.4390	0.0021	0.0041	0.0000
kelly_no_rebate@0.02	1000	HK$10,007,155	HK$9,938,587	0.4400	0.0043	0.0083	0.0000
kelly_no_rebate@0.05	1000	HK$10,013,567	HK$9,835,978	0.4590	0.0109	0.0206	0.0000
kelly_no_rebate@0.1	1000	HK$10,025,462	HK$9,673,977	0.4600	0.0217	0.0409	0.0000
kelly_settled_rebate@0.01	1000	HK$10,003,246	HK$9,969,687	0.4390	0.0021	0.0041	0.0000
kelly_settled_rebate@0.02	1000	HK$10,007,155	HK$9,938,587	0.4400	0.0043	0.0083	0.0000
kelly_settled_rebate@0.05	1000	HK$10,013,567	HK$9,835,978	0.4590	0.0109	0.0206	0.0000
kelly_settled_rebate@0.1	1000	HK$10,025,428	HK$9,673,827	0.4600	0.0217	0.0409	0.0000

Winners drawn from the model's own probabilities. 9 of 54 settled series shown, filtered to model view `bagged_offset`.

> **One simulation was declined rather than approximated**
>
> Refusals are recorded rather than omitted, so a reader can see the strategy was considered and declined. Rebate-aware sizing cannot reuse stale bankroll-dependent fractions on a divergent path.

The realised path, the day-block bootstrap and the model-assumed MC answer different questions and must not be shown as one uncertainty band.

> **What a simulation cannot tell you**
>
> The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

Technical appendix

CONFIGURATION

Item	Value
Market offset column	Implied_Prob

Item	Value
Features	28
Features the booster learns on	27
Opening bankroll	10,000,000
Bankroll mode	continuous
Rebate-aware sizing	no
Dataset registry digest	sha256:f93edc409844f3df108c777ef6f48931b6b0d095d5b9b16fb1ae1a590da08844

definition, raw_source, units, default_behavior, higher_value_meaning and known_caveats are intentionally blank. They are claims about the domain, not properties derivable from the model, and inventing them would produce documentation that reads as authoritative and is not.

KNOWN SOURCE-DATA DEFECTS

The lineage record carries `{'landing_dataset': 'aspx-results', 'landing_duplicate_keys': 1513, 'landing_primary_key': ['race_id', 'Horse No.'], 'retained_key': ['race_id', 'Horse No.'], 'retained_policy': 'keep_first', 'retained_policy_source': 'features/build.py', 'feature_table_duplicate_keys': 0, 'absorbed_by_build': 1513, 'consistent': True}`. Recorded here rather than corrected silently: the feature build absorbs these, and a reader auditing a number should know they exist.

Limitations and the retrospective-odds warning

> **Retrospective odds, restated**
>
> Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

> **Every test year is also development data**
>
> Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Bagged Offset (production) against the market on the same races. It does not measure this model against the other three except through that shared yardstick.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.